VR and Virtual Environments

Sunoikisis Digital Classics 2025–26: Digital Approaches to Cultural Heritage

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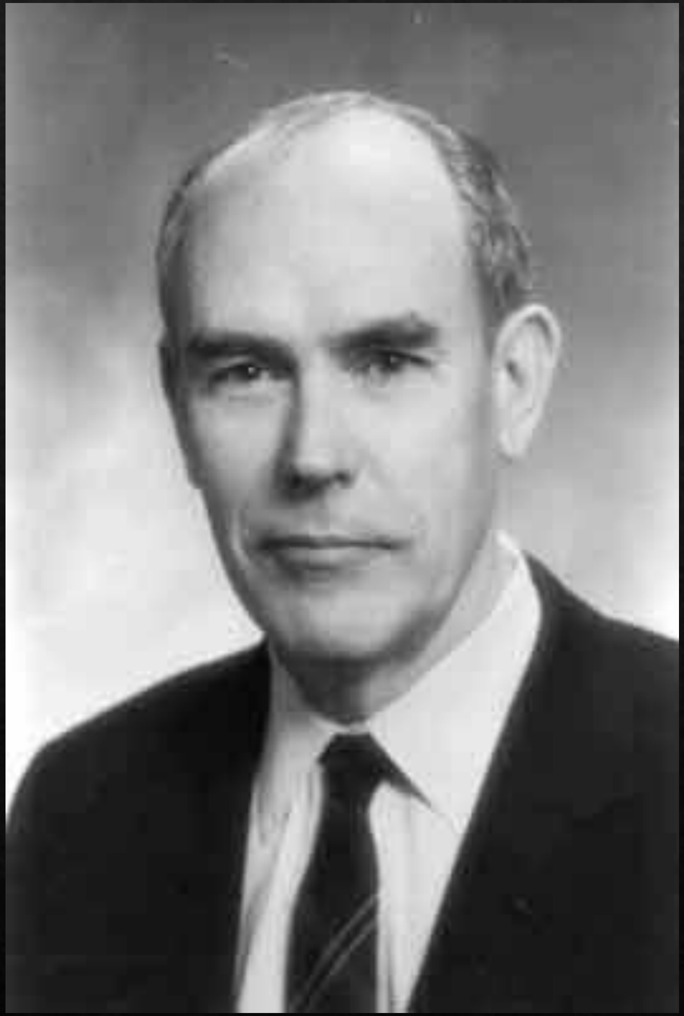
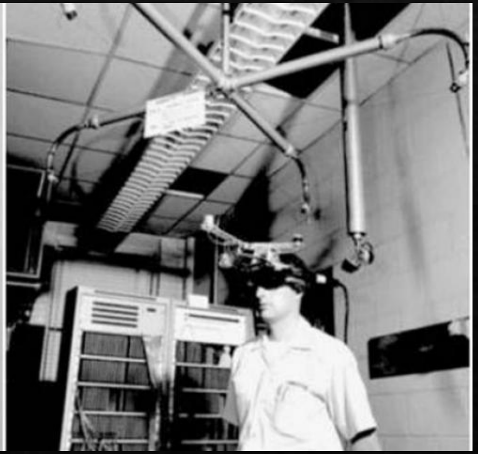
Session Overview

- Introduction to Virtual Reality
- VR for Cultural Heritage Collections & Sites
    - Theoretical Approaches
    - Case Study 1: The Sarapeion C on Delos
    - Case Study 2: the Swedish Pompeii Project
    - Case Study 3: Iconem
    - Case Study 4: Museums in the Metaverse
- Discussion & Questions


Virtual Reality (VR)

Pioneers: Ivan Sutherland





Pioneers: Jaron Lanier


VIRTUALITY

VIVE

htc | STEAM VR

VR: State of the Art

2016: affordable high-quality VR arrives

Oculus Rift S £300, Vive £500, Odyssey £500, Vive Pro £1,100

OK: needs £1k PC, £2k Laptop

Puzzle: Why do so few people have them?

2018: Standalone 3DoF headsets, early passthrough

Standalone headsets (Oculus GO) need no PC, no infrastructure.

But: Limited 3DoF experience

Passthrough allows us to see fuzzy representation of real world in black and white while onboarding

2025: Standalone 6DoF HMDs

Oculus Quest 3/S (£500/£300) defines the genre: no PC, no wires, no sensors but still 6DoF and full-colour passthrough.

Graphical trade-off, experience still excellent

Next: Better graphics/ergonomics

Medium term: 5G + Edge rendering

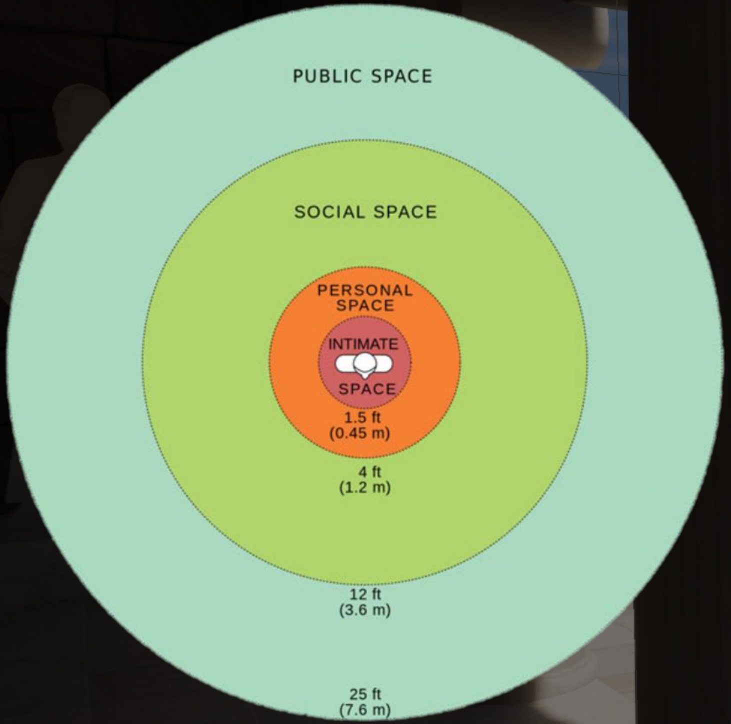
Long term: replaced by full-range XR devices that can be worn all-day

VR Features

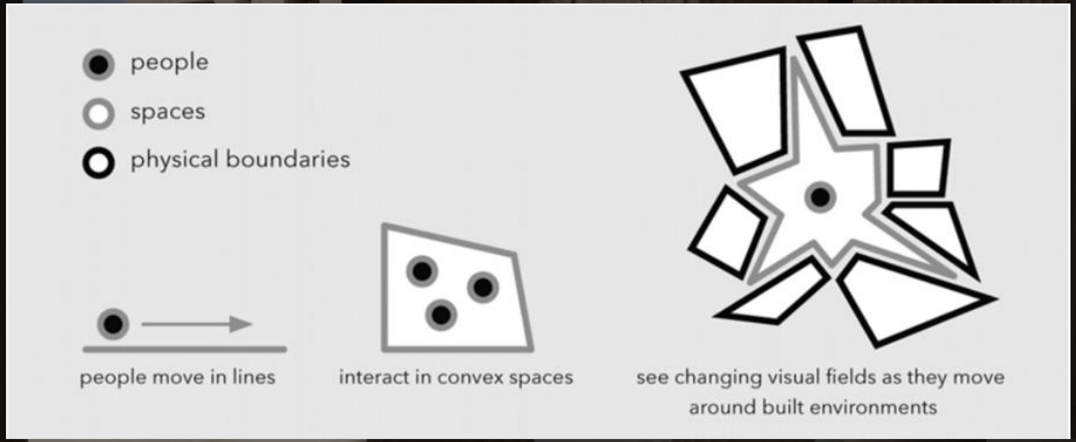
- Computer generated environments
- Sense of depth and 3D
- Immersive
- Isolating
- Disorienting


VR & Cultural Heritage

Theoretical approaches in archaeology: experiencing ancient spaces

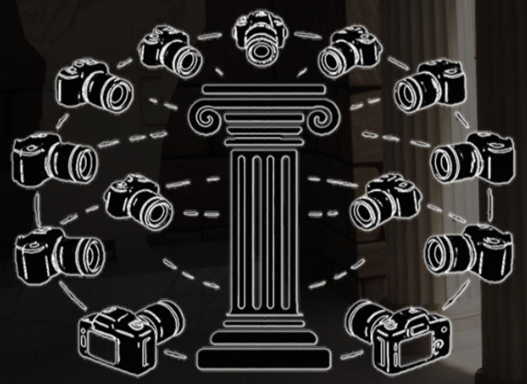


Hall's proxemics spaces as 'reaction bubbles' (Paliou et al. 2015: 125).



The structures of space according to human movements (Paliou et al. 2015: 20).

VR in Archaeology: between interpretation and experience



Photogrammetry → → 3D reconstruction

- Metashape
- Manual or automatic
- Capture every angle of an object or building

- Sketchup, Blender, etc.
- Manual reconstruction from data and interpretation
- Risk of speculation or errors

VR in Archaeology: between interpretation and experience

Data collection

- Data collected in the field (Raw data): GIS, plans photographs, drawings, measurements, etc.
- Publications (fieldwork, conference proceedings, books, articles, etc.).

Interpretation

- Main risks: speculation and/or errors
- But investigation and combination of evidence, experience and knowledge of larger trends.

Modelling

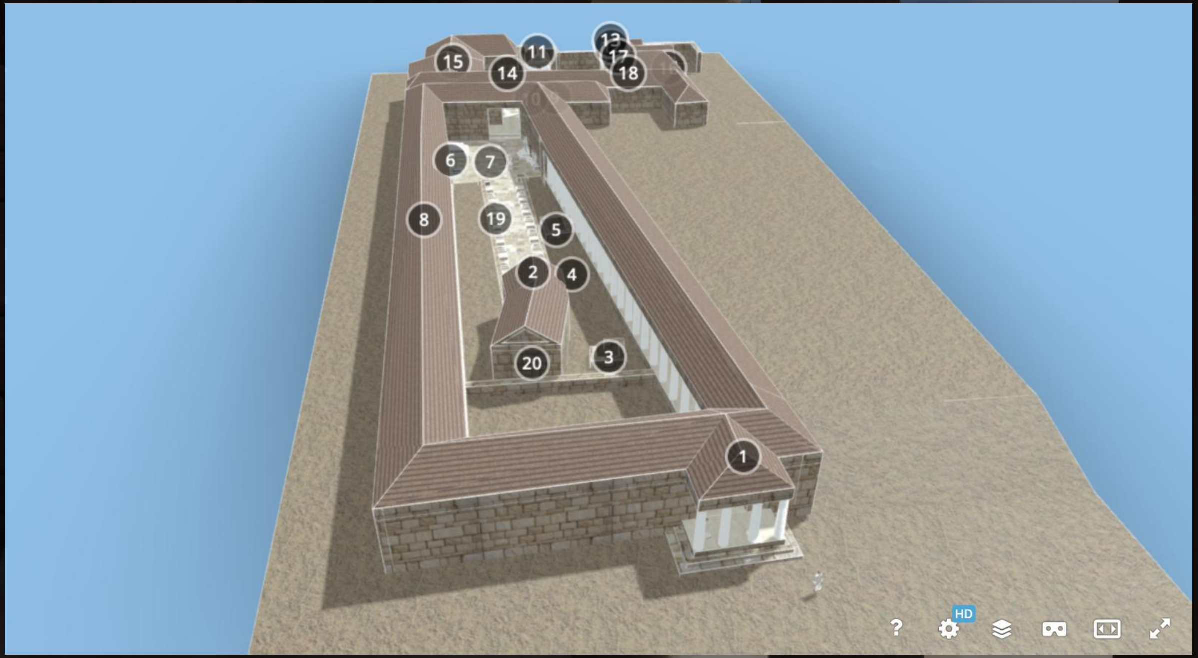
- Specialised software: Sketchup, Blender, Metashape, etc.
- Different types of models: photogrammetry, reconstruction, partial photogrammetry.
- Reconstruction of a built environment and its reality

Virtual Reality

- Visualisation.
- More interactive and autonomous approach.
- Accessibility of artefacts or buildings.
- Subjective interactions with the model.
- Teaching resource with different levels of experience.

Case study 1: the Sarapeion C on Delos

Sketchfab



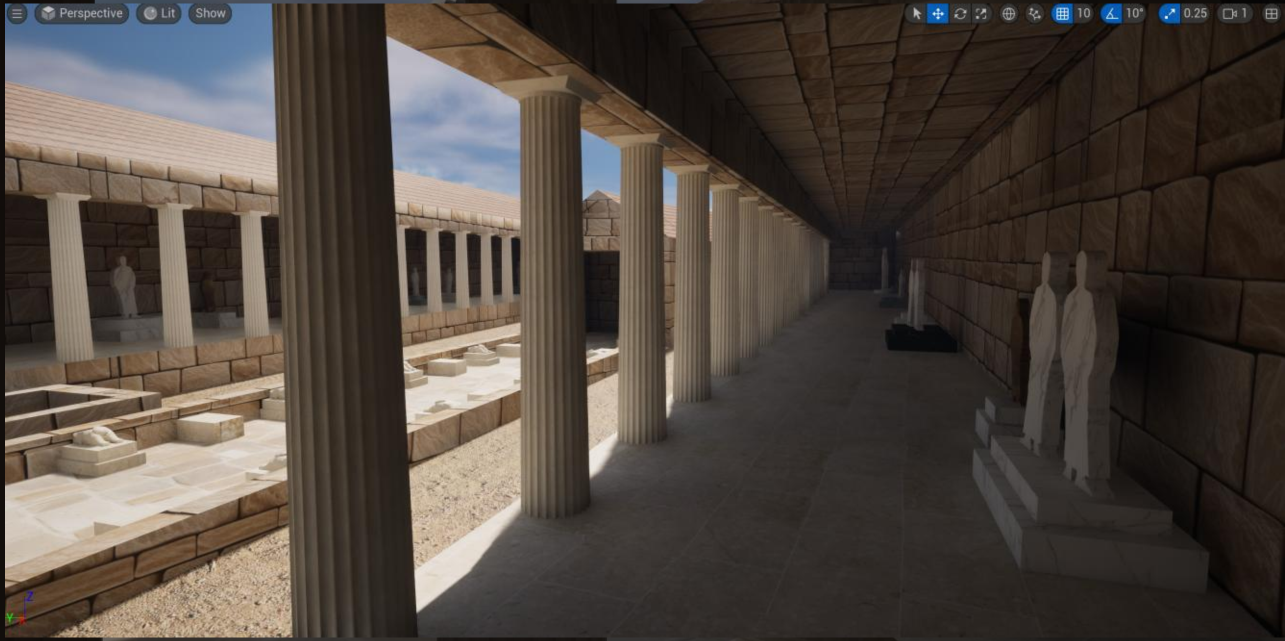
Unreal (game engine)

Sarapeion C: bird's eye view from the south

Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)

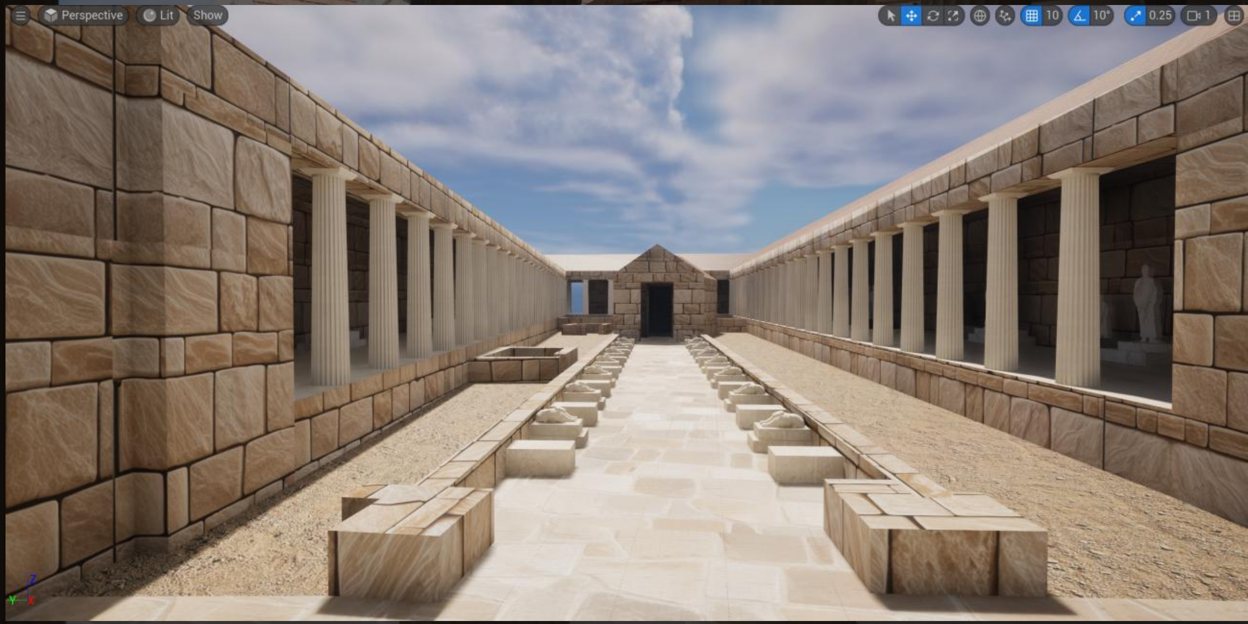


Sarapeion C: view from the east portico

Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)

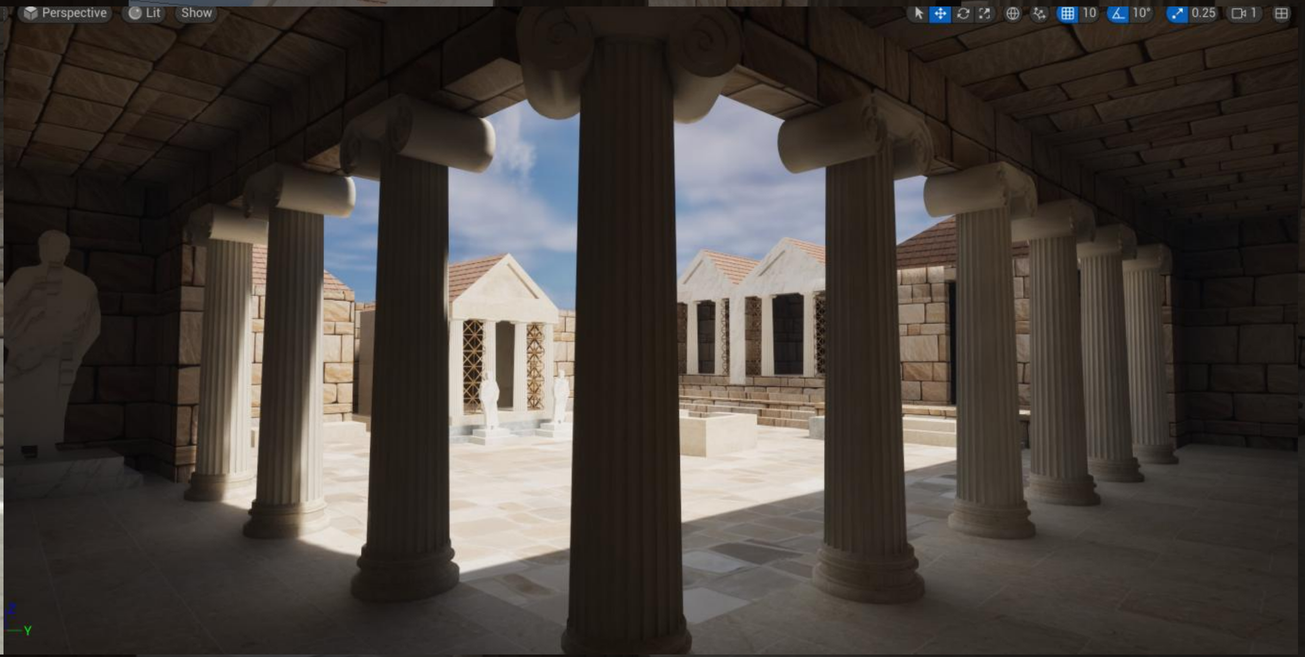


Sarapeion C: the dromos from the north

Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)

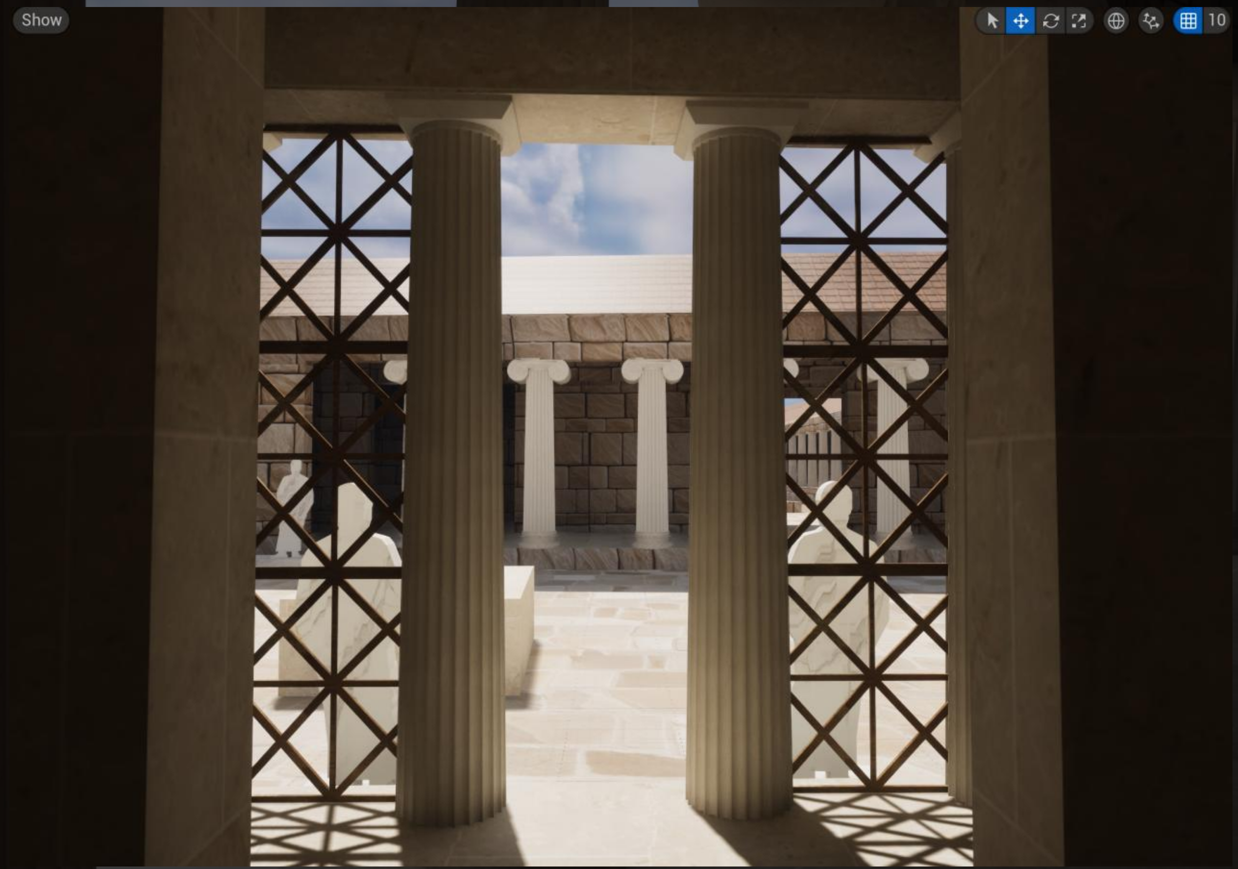


Sarapeion C: view of the northern courtyard and temples
from the L-shaped stoa

Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)



Sarapeion C: view on the northern courtyard
from the temple of Sarapis

Case study 2: the Swedish Pompeii Project

Lund University The Joint Faculties of Humanities and Theology Listen

The Swedish Pompeii Project

The joint faculties of humanities and theology

LUND UNIVERSITY

Project Insula V 1 Pompeii revived Search

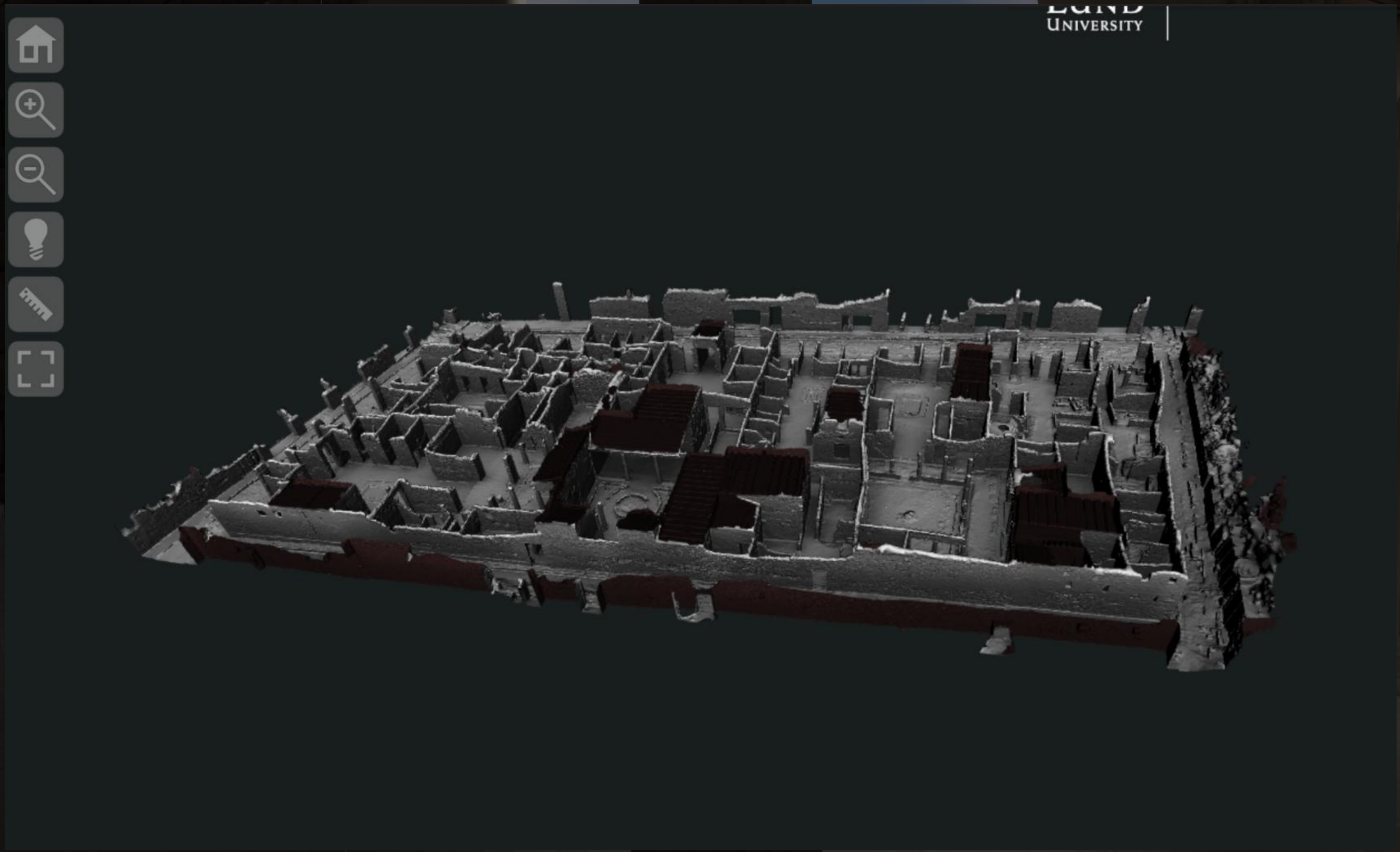
Start › Insula V 1 › Documentation of Insula V 1 › 3D Models & 3D GIS

- Introducing Insula V 1
- Documentation of Insula V 1
  - Archive Main Documentation
  - 3D Models & 3D GIS
  - Inscriptions of Insula V 1

3D Models & 3D GIS

Navigate the 3D-models by selecting a part of the insula. For best possible performance and compatibility, open the link with Google Chrome. (Try to reload the page if the 3D-model does not appear.)

Case study 2: the Swedish Pompeii Project

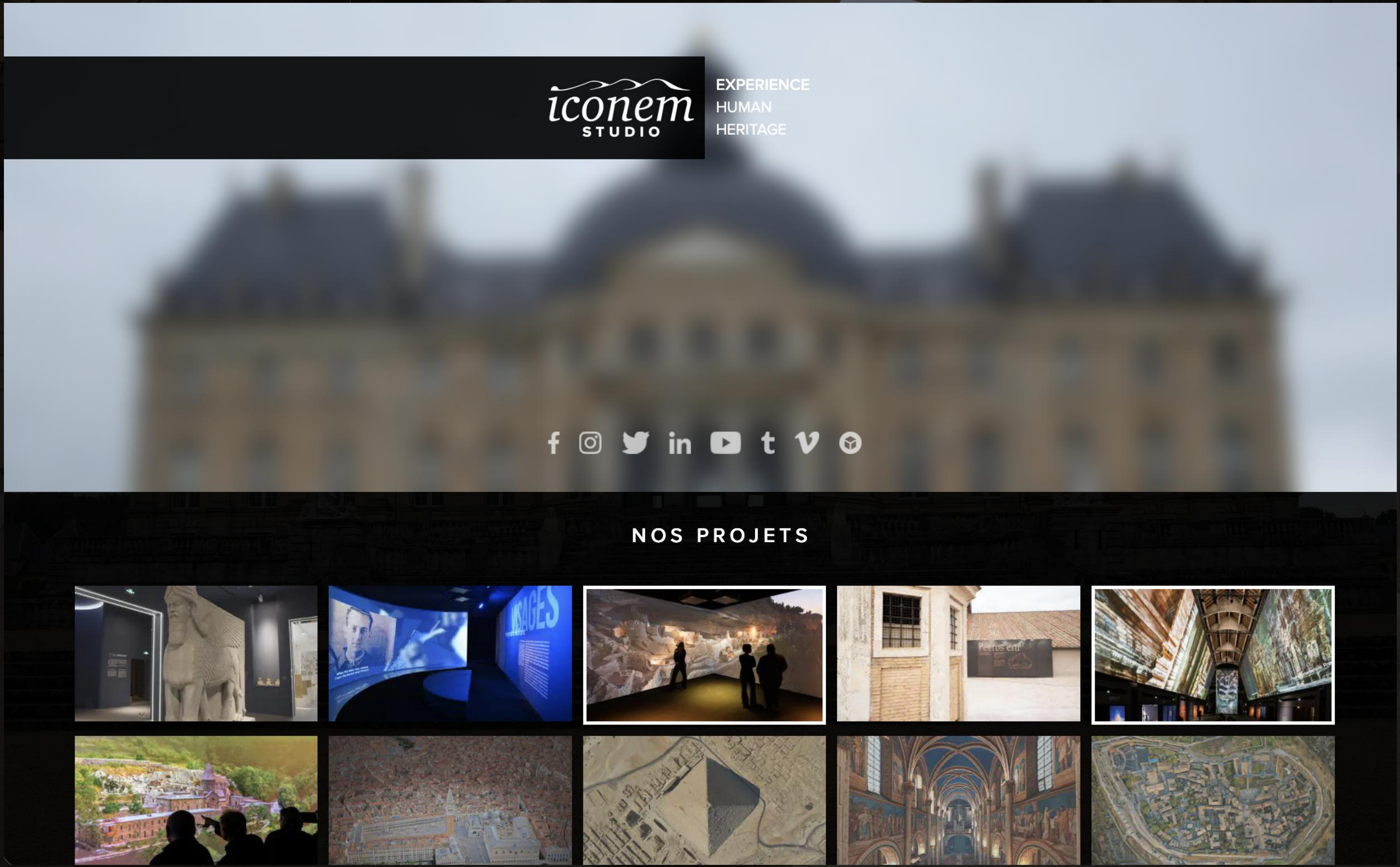


3D model of Insula V 1 (Swedish Pompeii Project)

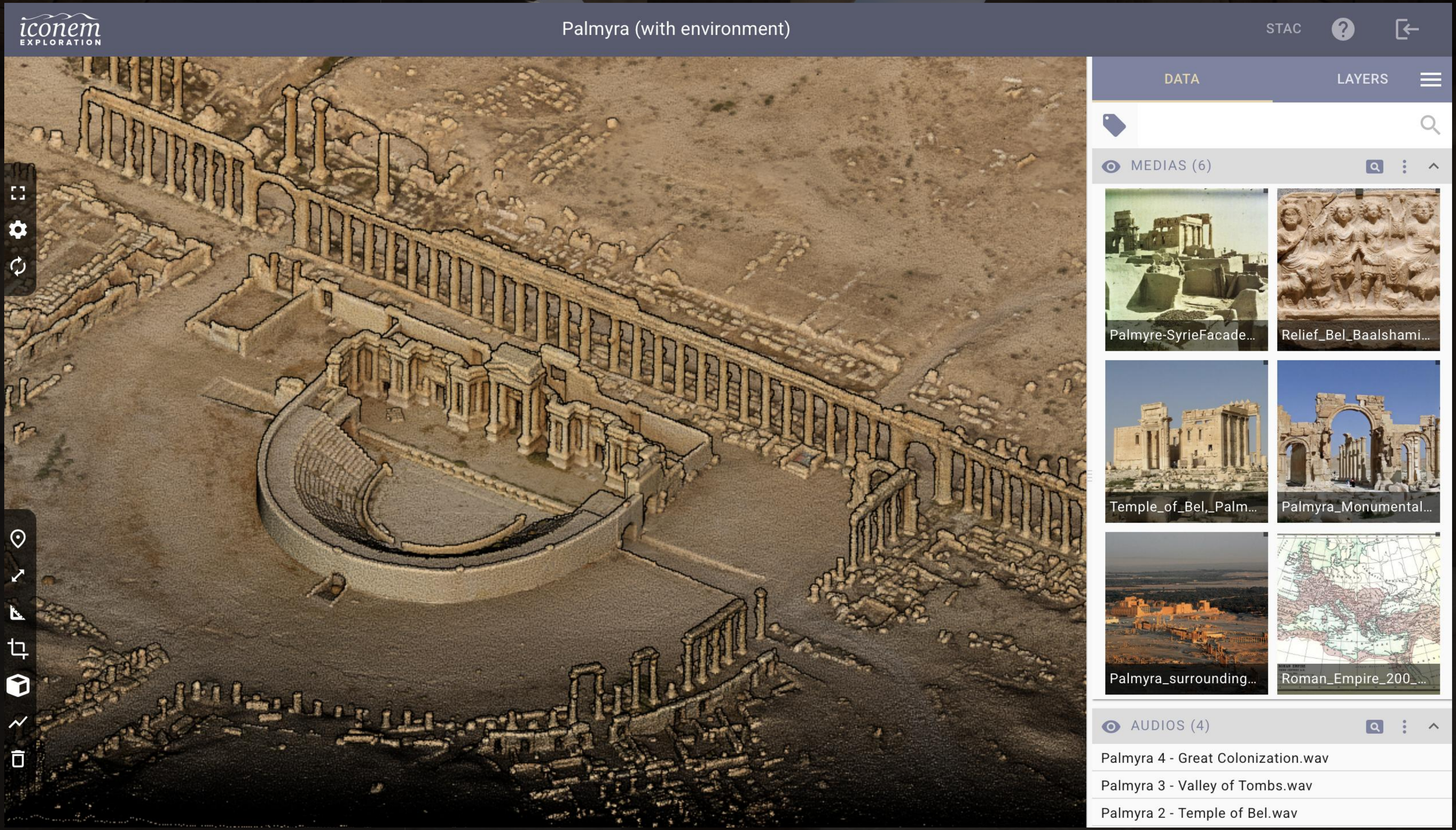
Case study 2: the Swedish Pompeii Project

Detailed models of buildings in Insula V 1
(Swedish Pompeii Project)

Case study 3: Iconem



Case study 3: Iconem - Palmyra



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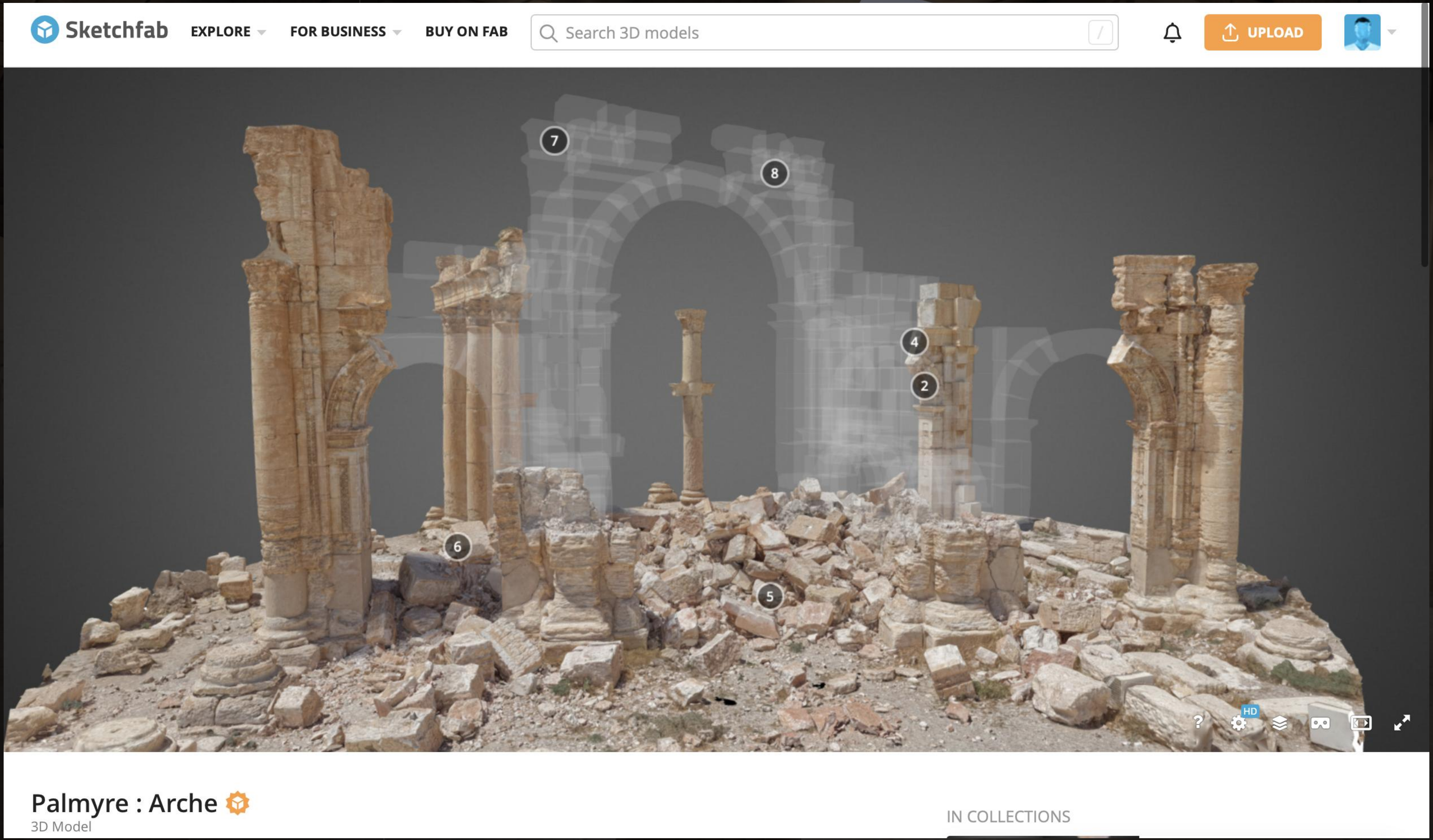
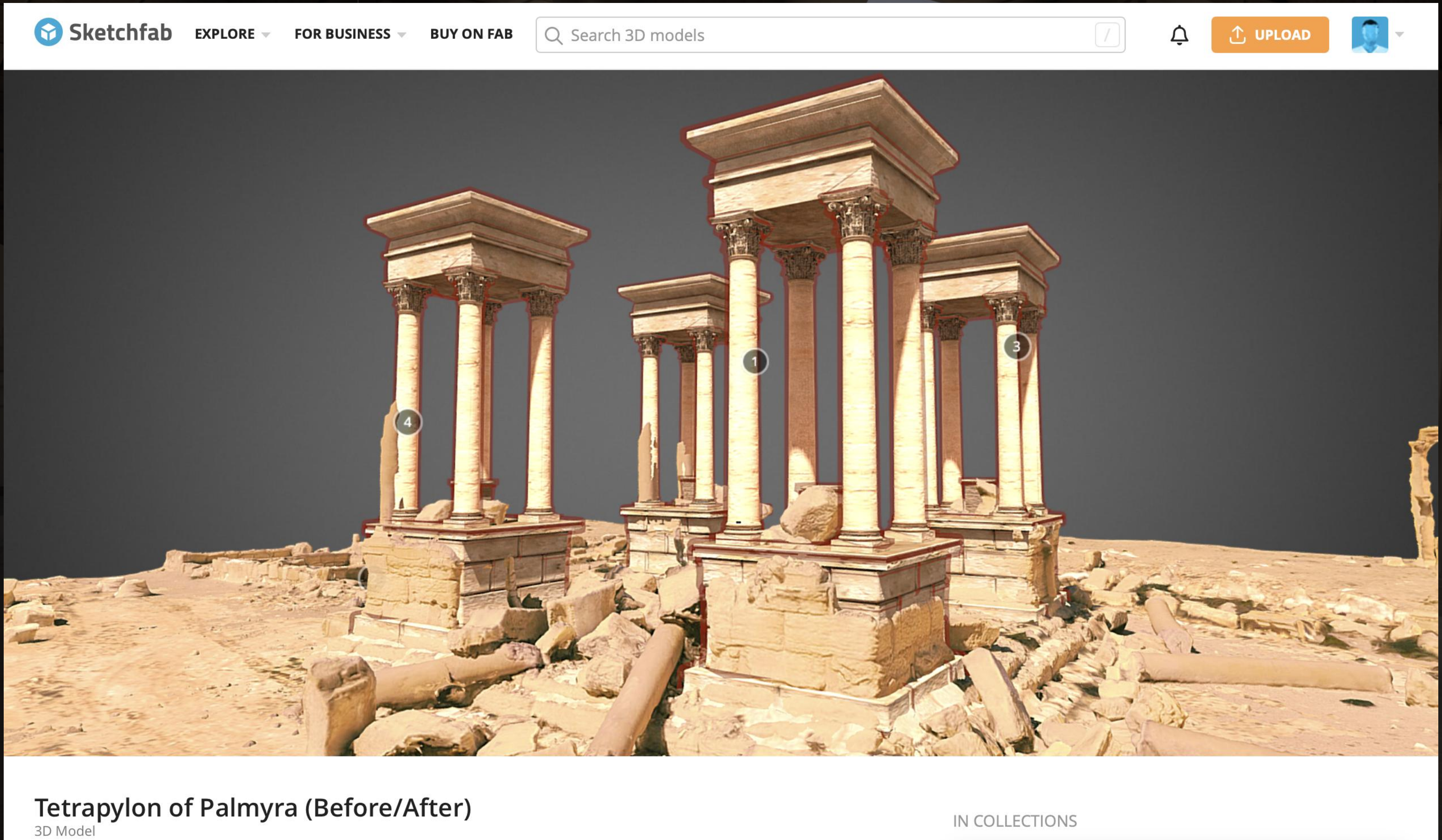


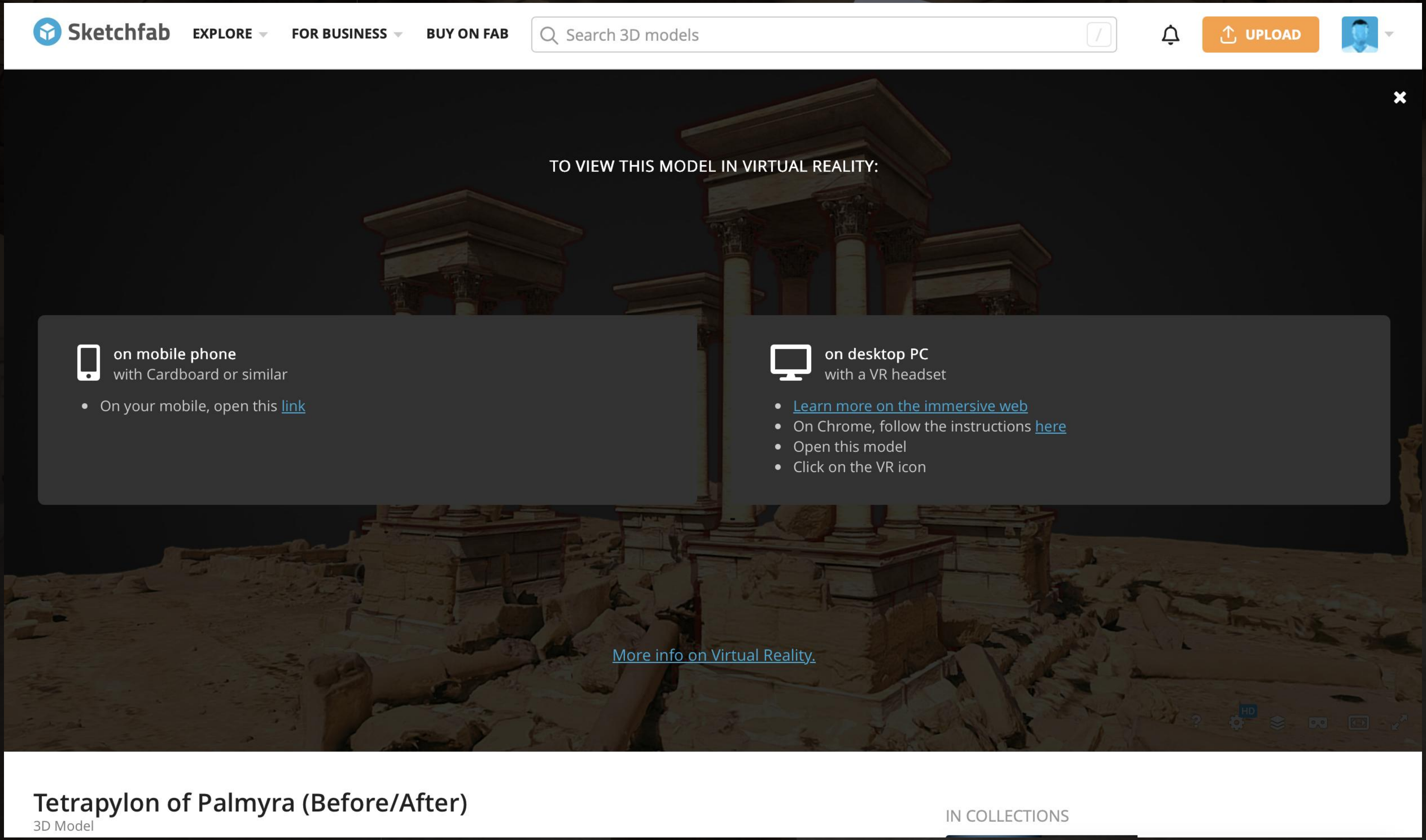
Screenshot of [Iconem on Sketchfab]: Arch in Palmyra with digital anastylosis

Case study 3: Iconem - Palmyra



Screenshot of [Iconem on Sketchfab](): 3D model of the Tetrapylon at Palmyra showing the site before and after destruction

Case study 3: Iconem - Palmyra



Screenshot of [Iconem on Sketchfab]: 3D model of the Tetrapylon at Palmyra showing the site before and after destruction

Museums in the Metaverse

Democratising XR for Cultural Heritage

UK Research and Innovation

Museums in the Metaverse

- £6.4M Levelling Up Innovation Accelerator Project (2023-26)
- Led by UofG in collaboration w/ Edify, The Hunterian, National Museums Scotland, Historic Environment Scotland (Phase 1)

- Project Aim: To make good on the economic promise of XR for the Cultural Heritage Sector

Museums

Huge economic and cultural importance, physically constrained.

Vast majority of objects in storage

Audience barriers

University of Glasgow

Credit: National Museums Scotland

Sensitive Objects

Virtual Museums

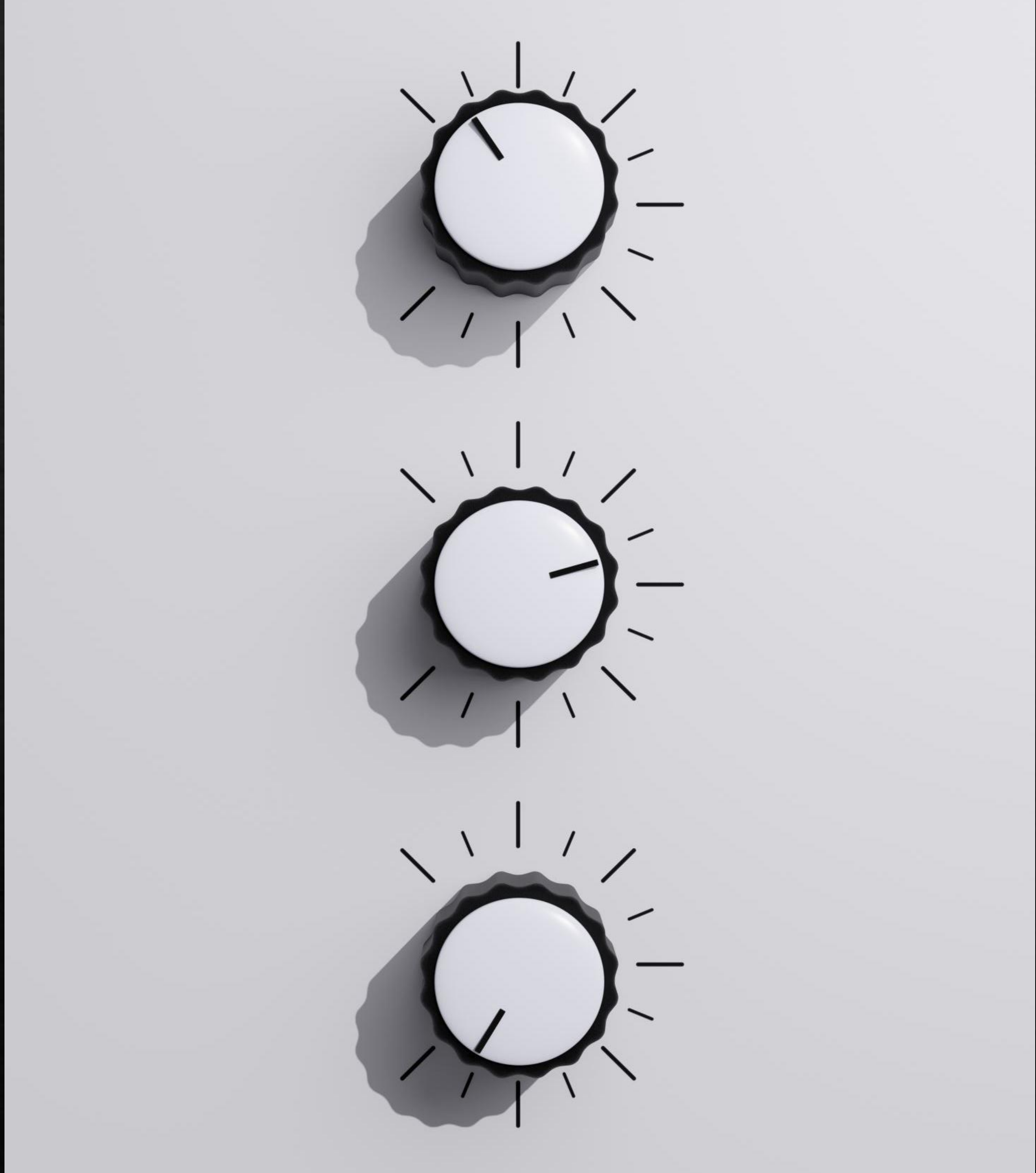
Obvious…. yet challenging.

Technology Barriers

University of Glasgow

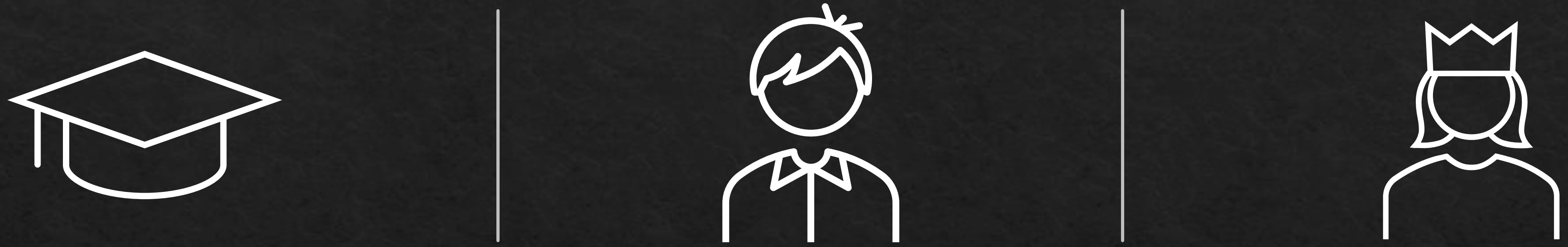
Economic Barriers

Control



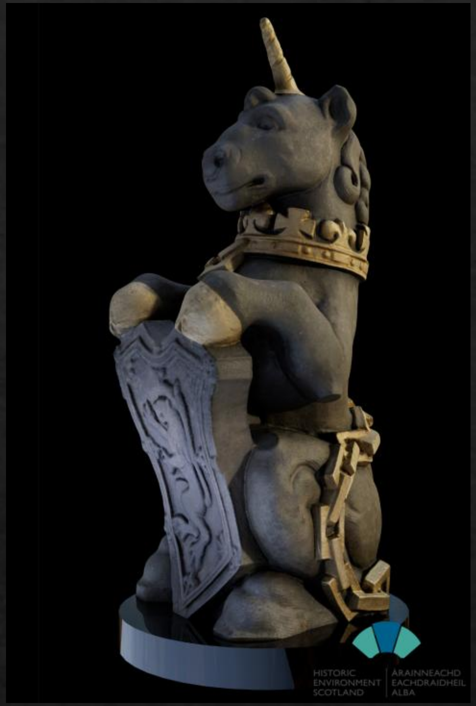
The MiM Solution

The core of the Museums in the Metaverse project.



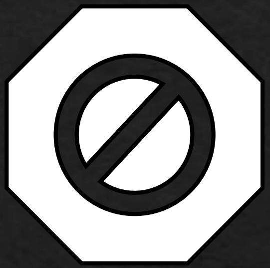
No Code Build-Your-Own (BYO) Museum Platform

* *Visitors: Access and Interact with curated MiM experiences on-site or remotely*
* *Creators: Heritage Professional, Lay Creator, Educator…*



BYO Museum with your own 3D assets or the heritage collections produced by MiM project team

User Generated Content (UGC)



IP, Availability, Veto

Controlled by object owner enabling *<u>safe digital loan</u>*

Cross-platform

Optimal in VR, available broadly (Desktop & Mobile)

Owners, Platform, Curators

Revenue shared fairly and sustainably

Experience heritage in captivating new ways:

- + Exquisite Object Detail
- + Impossible Interactions
- + Full Immersion
- + Rich Context(s)
- Collection-based Research & Teaching
- International Co-Curation

MiM in the Wild: [un]box

[un]box Deployment: Dec 2025 – Mar 2026

- Glasgow Science Centre, Glasgow
- The Hunterian, Glasgow
- St Giles' Cathedral, Edinburgh
- National Portrait Gallery, London

> **1,000+ user demos**
>
> **500+ in-person surveys**


Tools
Broadcast
Stop Broadcasting
00:00:00
Record Broadcast


Discussion & Questions

References

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Links

Swedish Pompeii Project

Iconem

Museums in the Metaverse